



THE UNIVERSITY OF
MELBOURNE

COMP90050 Advanced Database Systems

Semester 1, 2026

Lecturer: Farhana Choudhury (PhD)
Live lecture – Week 12

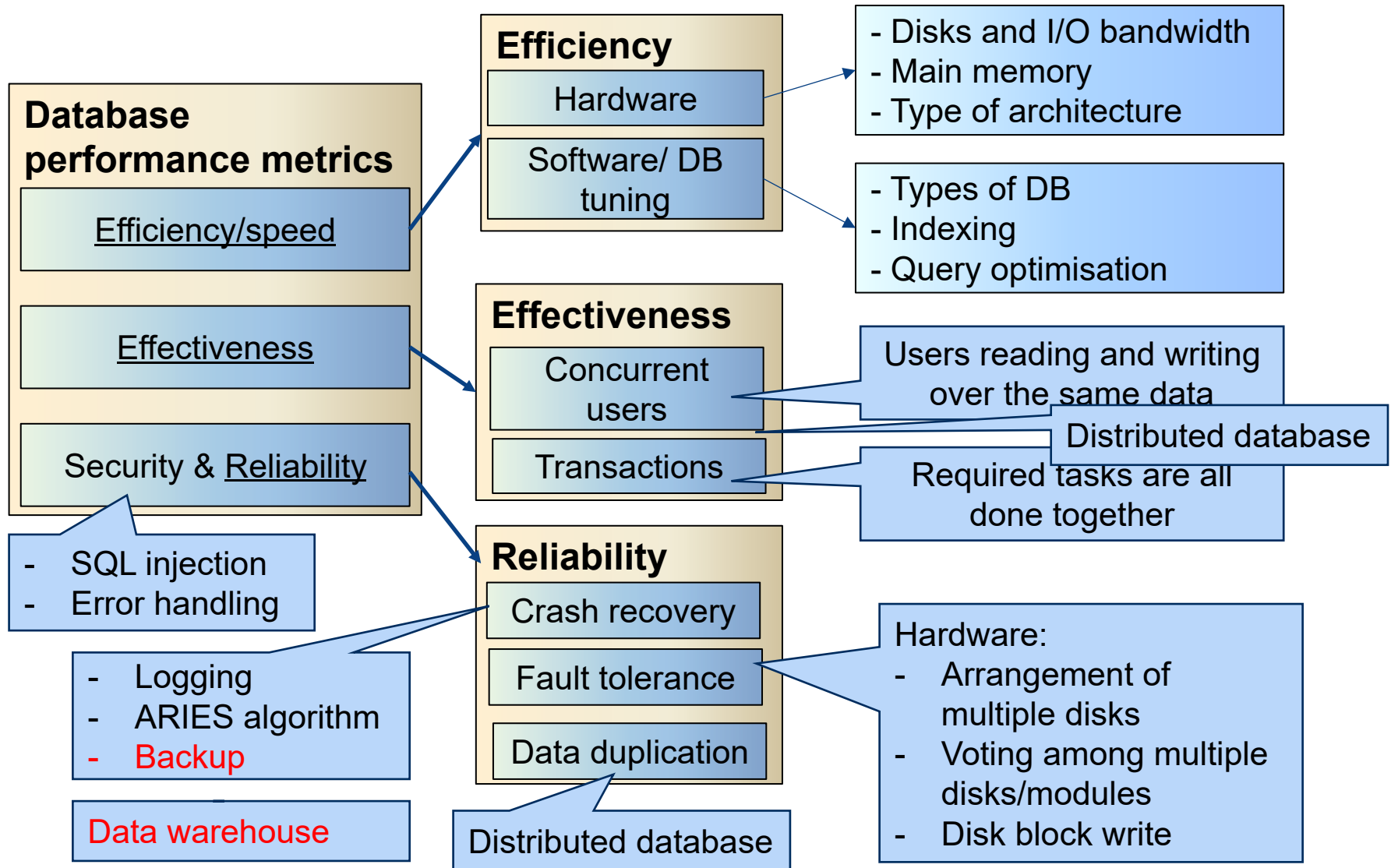




Today's topics

- Review this week's topics
- Recap of the subject
- Final exam preparation

Core Concepts of Database management system





Backup

News

Opinion

Sport

Culture

Lifestyle



The Guardian

Australia World AU politics Environment Climate crisis Indigenous Australia Immigration Media Business Health Science Tech Podcasts Newsletters

Reworked
Technology

Claude-powered AI agent's confession after deleting a firm's entire database: 'I violated every principle I was given'

PocketOS was left scrambling after a rogue AI agent deleted swaths of code underpinning its business

It only took nine seconds for an AI coding agent gone rogue to delete a company's entire production database and its backups, according to its founder. PocketOS, which sells software that car rental businesses rely on, descended into chaos after its databases were wiped, the company's founder Jeremy Crane said.

The culprit was Cursor, an AI agent powered by Anthropic's Claude Opus 4.6 model, which is one of the AI industry's flagship models. As more industries embrace AI in an attempt to automate tasks and even replace workers, the chaos at PocketOS is a reminder of what could go wrong.

Most viewed



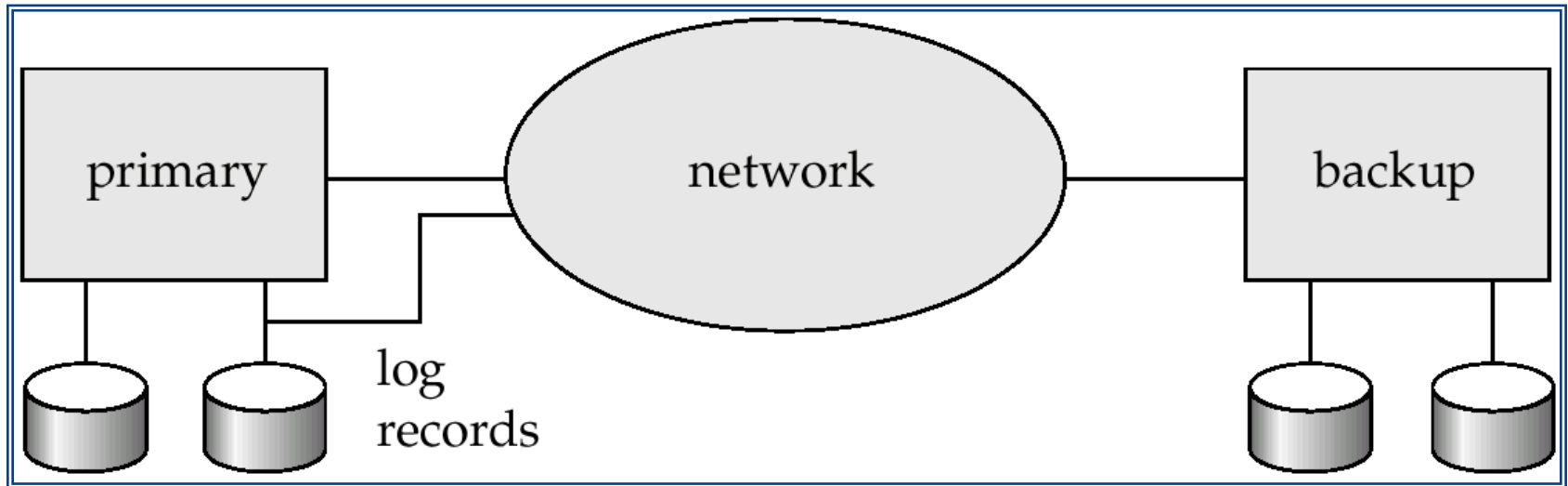
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Remote Backup Systems

Remote backup systems provide high availability by allowing transaction processing to continue even if the primary site is destroyed





We have also seen backups and recovery in practice

Strategy plan based on:

- Goals and requirement of your organization/task
- The nature of your data and usage pattern
- Constraint on resources

Design backup strategy:

- Full disk backup vs partial
- How frequently data changes
 - If frequent: use differential backup
- Space requirement of the backups – depends on the resource
- Multiple past instances of backup – useful if point-in-time recovery is needed



Crash recovery model (including backups) in practice

Choose the right recovery model for your application

Types of recovery models in MS SQL server:

- a. Simple: No logs, but has backups. Recovery is done from the last backup
- b. Full: Uses logs plus backups, regular checkpoints
- c. Bulk logged: Logs are not maintained for each individual writes, but for multiple writes together*

*For some operations



Data warehouse design issues

For what type of applications, simple recovery model can be useful/preferred over other recovery models?

Time for a poll –

[Pollev.com/farhanachoud585](https://pollev.com/farhanachoud585)





Crash recovery model (including backups) in practice

Choose the right recovery model for your application

1. Find the current recovery model

```
SELECT name, recovery_model_desc  
FROM sys.databases  
WHERE name = 'model' ;
```

2. Change the model if needed

3. **Checkpoint interval:** Specify target recovery interval, for example, if a crash happens, recover within 1 min. The system will then determine how often it needs to create checkpoints based on that value.



Crash recovery model (including backups) in practice

Diversify the point of failure – have genuine redundancy

UPDATED

AWS hit by US-East-1 outage after data center thermal event

News

May 11, 2026 • 5 mins

Amazon cloud unit AWS experiences major global outage disrupting hundreds of platforms

Internet

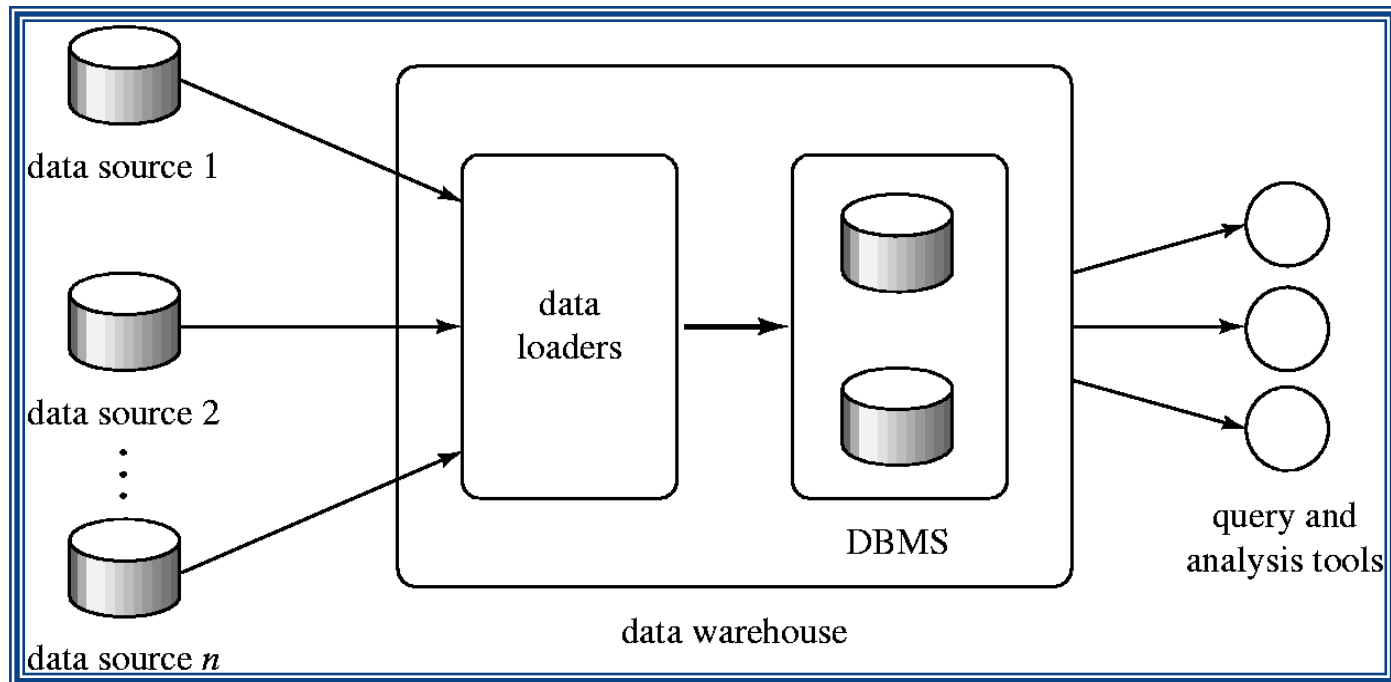
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Data warehouse design issues

When and how to gather data:

- **Source driven architecture:** data sources transmit new information to warehouse, either continuously or periodically (e.g. at night)
- **Destination driven architecture:** warehouse periodically requests new information from data sources





Data warehouse design issues

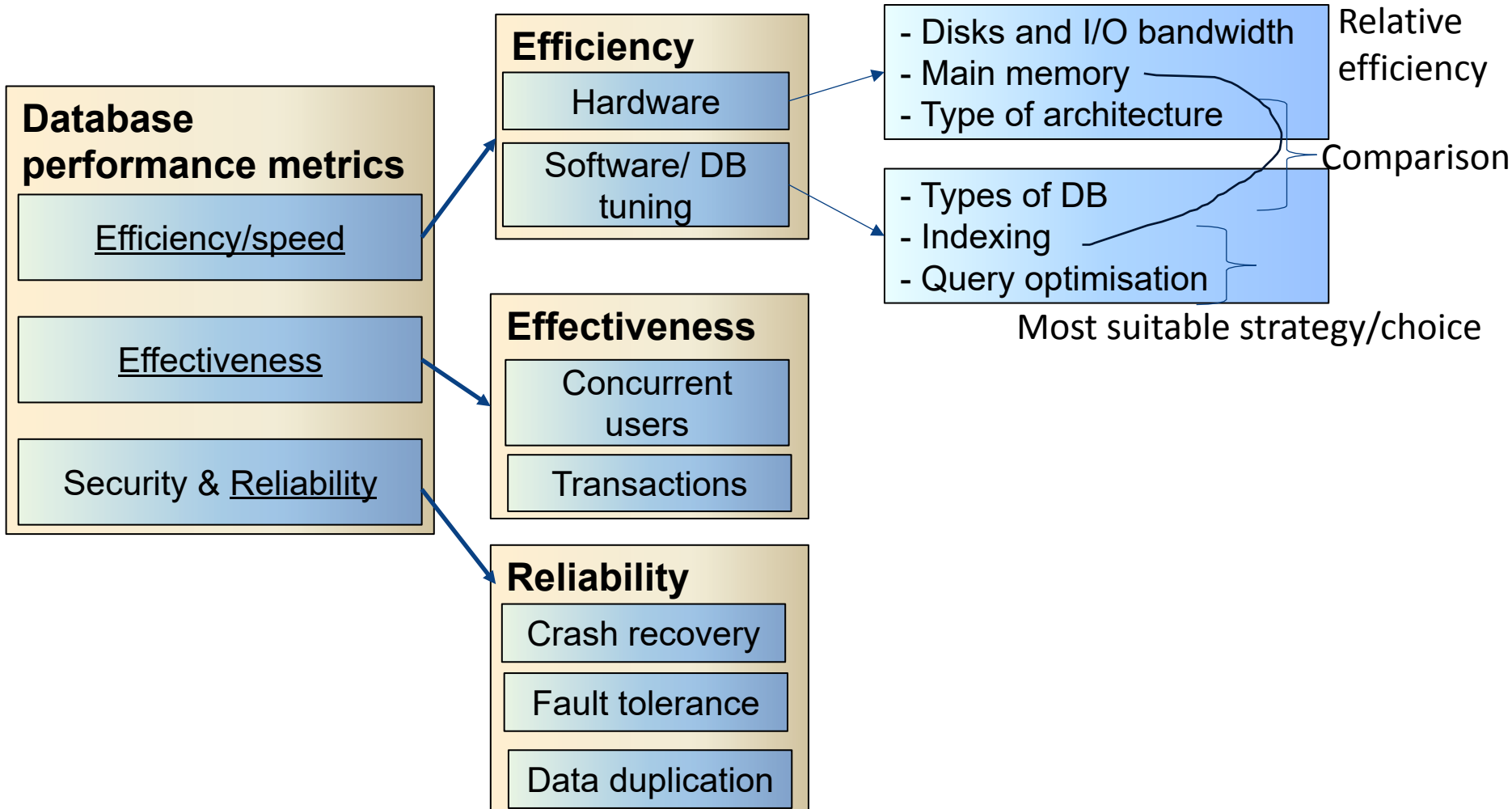
In a data warehouse, when should the data sources transmit new information to warehouse frequently, instead of periodically (e.g., at night)?

Time for a poll –

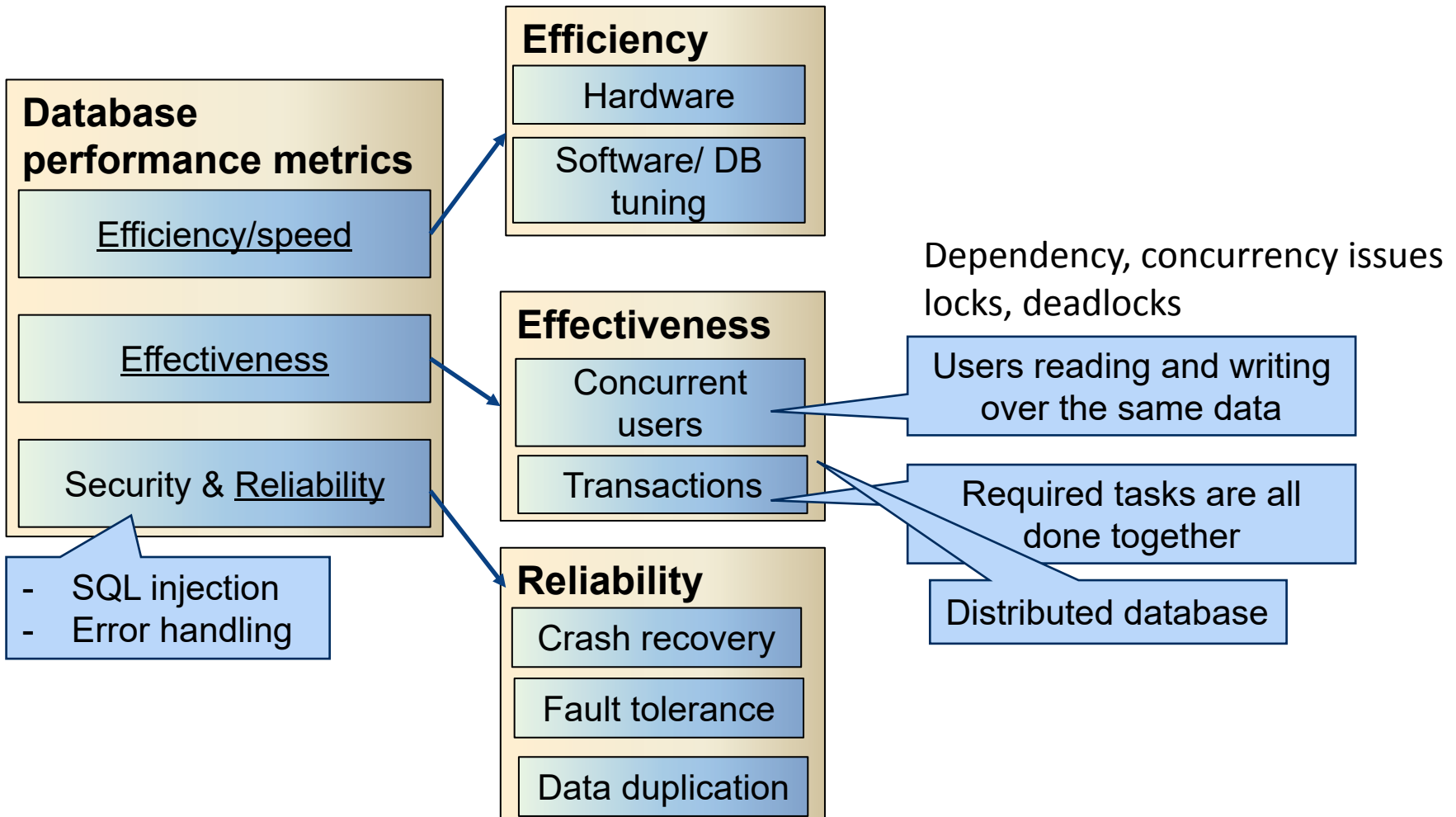
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Recap of the subject



Recap of the subject





Concept across topics

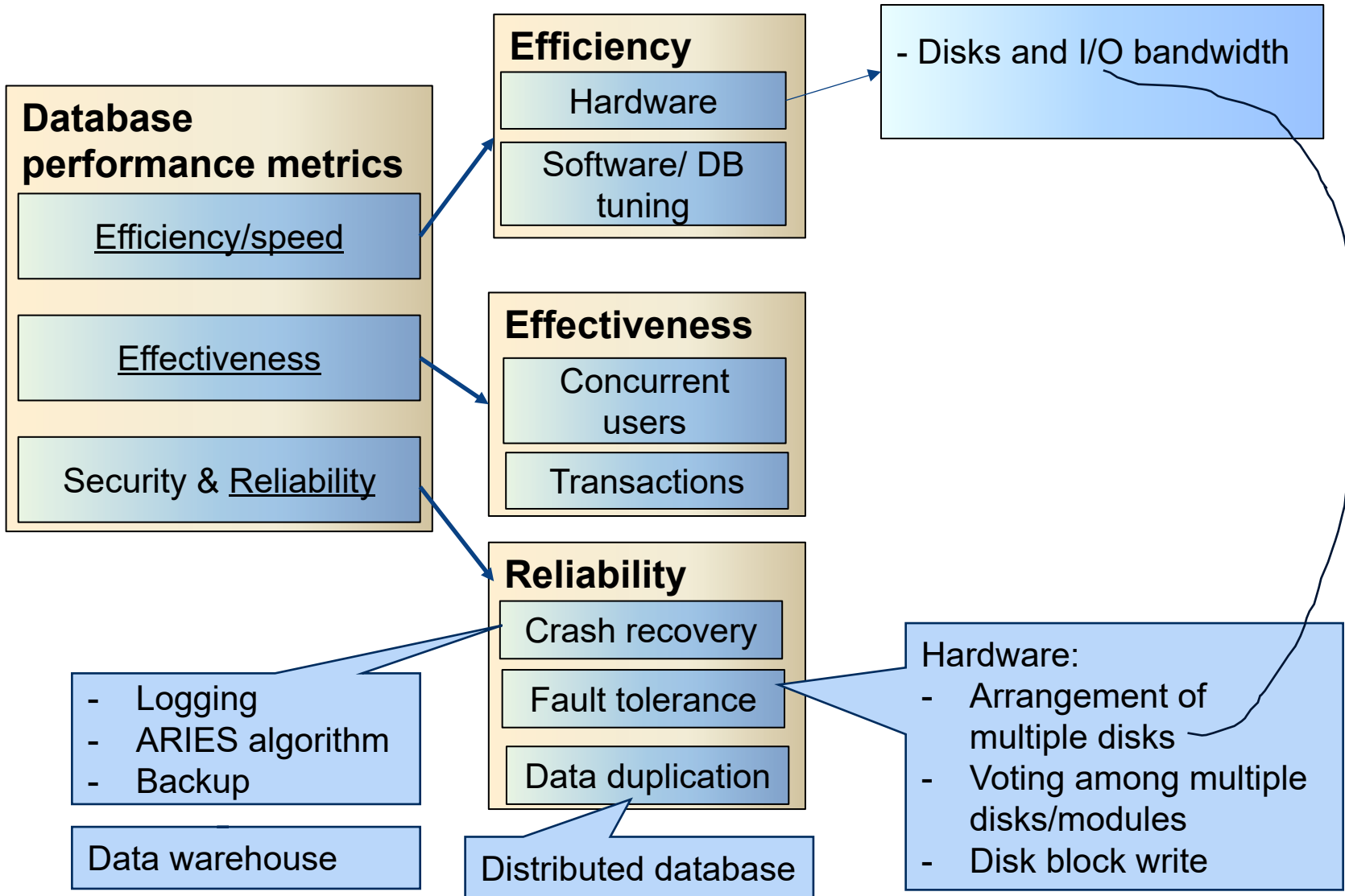
A question from live lecture 5 –

Can proper indexing reduce the chances of a deadlock?

A similar question in live lecture 7 –

Can optimistic locking reduce the chances of deadlock, compared to strict two-phase locking?

Recap of the subject



Example question

Consider the following relational database table. The most common queries on this table are – (i) Find all the postal addresses, and (ii) Find all the email addresses. However, the company that is running these queries finds that the query time is not as much fast as they would like to, especially when this relational database table has many records. An employee suggested that they can use columnar database to improve the efficiency of these two queries (without adding any new hardware). Is that correct? Explain the rationale.

ID	Name	Postal address	... (many other columns)	Email address
1	Smith	10 Central Street, Disneyland	...	smith@email.com
2	Zhou	22 South Road, Someland	...	zhou@email.com
3	Lin	12 North Road, Lalaland	...	lin@email.com



Final exam discussion

- Covering all topics (except lab materials)
- The exam is 2 hours long
- It is worth 50% of your total mark
- It's on-campus written exam
- It's open book - Student notes, printed materials, dictionaries - all are permitted
- Calculator is allowed
- You can use pencil to draw diagrams



Final exam discussion

- Read the questions!
- A question can have multiple parts (a, b, c ...)
- Different questions have different marks – answer accordingly



ESS now open

Use the following link for details -

<https://www.unimelb.edu.au/sls/students/ess>



THANK YOU!!!!!!